Light modes, moon texture data, and lithophane practice: research notes

Product context: a 50 mm 3D-printed lithophane moon, lit from inside by tunable-white LEDs (2700 K and 6500 K, CRI 90) plus an RGB LED. Prepared 2 Nov 2026.

How to read this file.

- Numbers marked **[computed]** were calculated in this session with CIE data. The scripts are in `/tmp/claude-0/-home-user/7ba8e87c-d4ee-5a66-9891-51bde4e0a9a7/scratchpad/scripts/`.
  - The CIE 1931 2° colour-matching functions come from colour-science 0.4.7.
  - The CIE S 026 melanopic and rhodopic action spectra come from the CIE S 026 toolbox v1.049, as bundled in luxpy.
- Numbers marked **[verified]** were read from the source page during this session.
- Numbers marked **[memory]** are from prior knowledge and were not re-verified this session.

> **Coverage gap.** The shared WebSearch budget for this turn (200 calls across all agents) ran out partway through. Section 3 (Thingiverse survey) is therefore **not done**; see §3 for what was checked and the follow-up plan. Sections 1, 2 and 4 used the searches done before the budget ran out, plus direct fetches of known URLs. No search engines were queried by other means, and no bot protection was bypassed.

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1. Light modes

1.1 Evidence base

Brown et al. 2022, PLOS Biology 20(3):e3001571 [verified]. This is the expert consensus, expressed in melanopic EDI (CIE S 026). All values are measured at the eye in the vertical plane, at about 1.2 m height.

Period	Recommendation (melanopic EDI)
Daytime	**≥ 250 lux** minimum. Spectrum ideally "enriched in shorter wavelengths".
Evening, starting ≥ 3 h before bed	**≤ 10 lux** maximum. White light should be "depleted in short wavelengths".
Night, sleep environment	"as dark as possible"; ambient **≤ 1 lux** maximum.
Night, if vision is needed	**≤ 10 lux** maximum.

Further points from the paper [verified]:

- Common domestic warm-white (2700–3000 K) LEDs already have a melanopic DER "sufficiently low (<0.35…)" to meet the 10 lux evening limit at about 30 lux photopic.
- Ultra-low-melanopic sources "will likely come at the expense of reduced colour discrimination".

Converting to photopic lux. melanopic EDI = melanopic DER × photopic lux. So the most photopic lux a mode can deliver is 10 / DER (evening) or 1 / DER (sleep).

Children.

- Children show greater melatonin suppression than adults at both 3000 K and 6200 K. The authors recommend low-CCT light at night (Lee et al. 2018, Physiol Rep, PMC6295443) [verified, from a search snippet].
- Glickman et al. 2026 (npj Biological Timing and Sleep 3) measured 25 top-rated children's nightlights across 79 settings [verified, from a search snippet and the USUHS listing].
    - Only 3 of the 25 stayed below 5 lux melanopic EDI at the source, and 2 of those met the adult limit of < 1 lux.
    - "The vast majority do not even meet evening and nighttime recommendations for adults."
    - Their advice is to make nightlights "**dim, red, and far from the bed**".

Sleep Foundation (updated 8 Sep 2026; medically reviewed by P. Adavadkar MD) [verified]:

- "A dark room is best for sleep."
- "Red light does not affect circadian rhythms…"
- "Dim yellow and orange colored lights have little impact…"
- It warns that red light that is too bright can still suppress melatonin.
- A Cleveland Clinic specialist (WBRZ news piece) adds that young children may find red scary, so orange or amber is an alternative [verified].
- **AAP:** no official AAP or HealthyChildren.org night-light colour statement was found before the search budget ran out. Do not cite the AAP for colour advice until one is confirmed.

Dark adaptation [computed]. Rod (rhodopic) DER, where D65 = 1:

Source	Rhodopic DER
625 nm red	0.018
630 nm red	0.015
660 nm red	0.006
590 nm amber	0.094
2700 K	0.54
6500 K	0.99

Red at 620–630 nm therefore barely stimulates rods, which is why it preserves dark adaptation. Deep red at 660 nm is better still but looks dimmer per watt.

Flicker: IEEE 1789-2015 [verified via the DigiKey and DOE summaries]. The IEEE site now lists the standard as "Inactive-Reserved", but it is still the usual reference.

Level	< 90 Hz	Middle band	No limit above
Low risk: maximum Mod%	0.025 × f	0.08 × f, from 90 to 1250 Hz	1250 Hz
No observable effect: maximum Mod%	0.01 × f	0.0333 × f, from 90 to 3000 Hz	3000 Hz

Firmware recommendation:

- Run the LED PWM at **≥ 3 kHz**, so that 100 % modulation is exempt. Ideally use ≥ 20 kHz to avoid audible coil or MLCC noise and camera banding.
- Use ≥ 12-bit resolution, plus a gamma curve, so that 1–3 % night levels dim smoothly.
- Constant-current analog dimming is an alternative at the lowest levels.

1.2 Melanopic DER and maximum lux per mode [computed]

The white values below are for **blackbody** spectra. Real CRI-90 phosphor LEDs usually come out about 5–15 % lower at 2700 K. Brown et al. quote 0.83 for a 6500 K tunable LED. Recompute these with your LED datasheet's SPD.

Source	mel DER	Max photopic lux for 10 mel-EDI (evening)	Max photopic lux for 1 mel-EDI (sleep)
1600 K	0.189	53	5.3
1800 K	0.241	42	4.2
2200 K	0.343	29	2.9
2700 K	0.462	22	2.2
3000 K	0.526	19	1.9
4000 K	0.708	14	1.4
5000 K	0.845	12	1.2
6500 K	0.993	10	1.0
Red 625 nm (Gaussian, 17 nm FWHM)	0.0026	~3900	~390
Amber 590 nm direct emitter	0.020	~500	~50
PC-amber (broad)	0.081	~124	~12
Green 525 nm	0.888	11	1.1
Blue 465 nm	10.8	0.9	0.09

Key finding for this LED set (2700 K + 6500 K + RGB) [computed]. A candle or amber colour mixed from **red plus a 525 nm green is not low-melanopic**, because green carries about as much melanopic stimulus per lux as daylight.

Mix	mel DER
R + G "amber" (dominant 590 nm)	0.40 (direct 590 nm amber: 0.02)
1800 K from 2700 K + R + G (22 / 45 / 33 % luminance)	0.39 (blackbody: 0.24)
2200 K from 2700 K + R + G (53 / 25 / 21 %)	0.44
Red + 2700 K only, no green: 70 % red luminance (dominant 612 nm)	0.14
Red + 2700 K only, no green: 50 % red luminance (605 nm)	0.23
Red + 2700 K only, no green: 20 % red luminance (≈1870 K, Duv −0.009, slightly pink)	0.37

Conclusions:

- The **night modes should use red, or red plus a little 2700 K, and never green.**
- A genuinely low-blue amber needs a **dedicated 590 nm amber or PC-amber LED**. This is worth considering for the board.
- Candlelight from this LED set mostly wins because it is **dimmer**, not because of its spectrum.

1.3 Recommended modes (11), in cycling order from night to day

Notes on the table:

- "Default" is the suggested power-on level for the mode, as a percentage of that mode's maximum.

- The eye-lux target is the illuminance at the user's eye at about 0.5–1 m.
- The modes are in the order a user would cycle through them.

#	Name	Target	Duv	Default	Purpose	Evidence / notes
1	**Night red**	Red LED alone, dominant ≈ 620–625 nm, xy ≈ (0.696, 0.304)	n/a	2 %, ≤ 0.5–1 lux at pillow	All-night and middle-of-the-night light; preserves dark adaptation	mel DER 0.003, rhodopic DER 0.018. Brown: ≤ 1 lux mel-EDI. Glickman: "dim, red, far from bed". Sleep Foundation: red is the best choice.
2	**Soft orange / amber night**	Red + 2700 K at about 70 / 30 % luminance, dominant ≈ 610–612 nm. With an amber LED: 590 nm.	n/a	3 %	Night light for children who dislike red	mel DER 0.14 (dedicated amber: 0.02). Do **not** mix it from R + G (0.40).
3	**Candlelight**	1800–1900 K. 1800 K is x 0.5492, y 0.4082.	0 to +0.002	10 %	Wind-down in the last hour before bed	Mixed as 2700 K + R + G, so mel DER ≈ 0.39. Keep ≤ 25 lux at the eye to stay under 10 mel-EDI.
4	**Warm white** ("yellow / natural white")	2700 K, x 0.4599, y 0.4106 (the 2700 K channel alone)	0	35 %	Main evening ambient light	mel DER ≈ 0.40–0.46. ≤ 22 lux at the eye in the evening.
5	**Soft white**	3000 K, x 0.4369, y 0.4041 (about 87 % 2700 K, 10 % 6500 K, 2.5 % green)	0	50 %	Evening reading	mel DER ≈ 0.53. ≤ 19 lux at the eye in the evening.
6	**Neutral / reading**	4000 K, x 0.3805, y 0.3768 (about 53 % 2700 K, 42 % 6500 K, 5 % green)	0, ANSI nominal +0.001	100 %	Daytime reading and task light	mel DER 0.71. Above the evening limit beyond ~14 lux at the eye, so present it as a **day** mode.
7	**Daylight**	5000 K, x 0.3451, y 0.3516 (about 28 % 2700 K, 69 % 6500 K, 3.7 % green)	0 / +0.002	100 %	Daytime work and colour checking	mel DER 0.84
8	**Cool / blue-white**	6500 K, x 0.3136, y 0.3237 (the 6500 K channel alone)	0 / +0.003	100 %	Morning and daytime alertness only	mel DER ≈ 0.83–0.99. Brown: daytime ≥ 250 mel-EDI, which the lamp can't reach, so treat it as ambience. Cap it in the evening.
9	**Moonglow**	≈ 4100 K (the real colour of moonlight), at a very low level	0	3 %	Decorative "real moon" glow; auto-off timer	Moonlight measures ≈ 4000–4250 K and only *looks* blue (Purkinje shift). Full moon gives ≈ 0.05–0.1 lux, at most ≈ 0.3 lux (Kyba et al. 2017). A blue "moonlight" (≥ 8000 K, mel DER ≥ 1.1) is common in night-light products but wrong for sleep. If offered, keep it decorative and time-limited.

#	Name	Target	Duv	Default	Purpose	Evidence / notes
10	**Green**	RGB green, ≈ 525 nm	n/a	20 %	Decorative or indicator colour (owner request)	mel DER 0.89, about the same as daylight per lux. Not a night mode.
11	**Blue**	RGB blue, ≈ 465 nm	n/a	10 %	Decorative (owner request)	mel DER ≈ 10.8: 1 lux of blue equals about 10 mel-EDI lux, the whole evening budget. Firmware should block or cap it at night.

Optional extra: **Sunset 2200 K** (x 0.5056, y 0.4152). It gives almost no melanopic benefit over 2700 K with this LED set (0.44 vs 0.46), so merge it into Candlelight unless an amber LED is added.

Naming caution. In US lamp marketing, "Soft White" usually means 2700 K and "Warm White" 3000 K, the opposite of the labels used here. Show the CCT in the app or on the label.

Reading level [memory]. Typical reading recommendations (IES, EN 12464-1) are about 300–500 lux on the page.

- A 50 mm lithophane sphere with an output of tens of lumens will give only tens of lux at 0.4–0.5 m. This is my estimate, not measured.
- So "reading" here means bedside, low-level reading. State that in the product copy.

Firmware idea.

- Store a per-mode melanopic DER, measured from the real LEDs and recomputed for each mix.
- Estimate melanopic EDI at a nominal 0.5 m from the drive level.
- In an "evening" time window, cap brightness to ≤ 10 mel-EDI. In a "night" window, cap it to ≤ 1 mel-EDI.

1.4 Planckian chromaticities (CIE 1931 2°) for firmware [computed]

The "exact" columns were integrated from the CIE 1931 2° colour-matching functions at 380–780 nm in 1 nm steps, with $c_2 = 1.4388 \times 10^{-2}$ m·K. They were cross-checked against Kim et al. 2002 (cubic spline) and Krystek 1985 (rational u,v); all agree to within about 0.0006.

CCT	x (exact)	y (exact)	x (Kim)	y (Kim)	u′	v′
1600 K	0.5732	0.3993	(outside range)	(outside range)	0.3451	0.5408
1800 K	**0.5492**	**0.4082**	0.5496	0.4081	0.3231	0.5403
1900 K	0.5378	0.4112	0.5381	0.4111	0.3136	0.5396
2200 K	**0.5056**	**0.4152**	0.5054	0.4152	0.2901	0.5360
2700 K	**0.4599**	**0.4106**	0.4593	0.4107	0.2625	0.5274
3000 K	**0.4369**	**0.4041**	0.4366	0.4042	0.2506	0.5214
3500 K	0.4053	0.3907	0.4053	0.3909	0.2357	0.5113
4000 K	**0.3805**	**0.3768**	0.3805	0.3767	0.2251	0.5016
5000 K	**0.3451**	**0.3516**	0.3450	0.3516	0.2114	0.4847
6500 K	**0.3136**	**0.3237**	0.3135	0.3237	0.2004	0.4656

- For reference, the CIE daylight-locus points are D50 (0.3457, 0.3586) and D65 (0.3127, 0.3291). Both sit above the Planckian locus.
- Kim et al. 2002 cubic spline, valid for 1667–25000 K:
  - $x = -0.2661239e9/T^3 - 0.2343589e6/T^2 + 0.8776956e3/T + 0.179910$ for $T \le 4000$ K
  - $x = -3.0258469e9/T^3 + 2.1070379e6/T^2 + 0.2226347e3/T + 0.240390$ above 4000 K
  - y is then a cubic in x, with coefficient sets for three ranges: up to 2222 K, 2222–4000 K, and above 4000 K. The code is in `colour_calc.py`.

Mixing 2700 K and 6500 K [computed]. A mix of two whites lies on the straight line between them in xy (and in u′v′). That line passes *below* the Planckian locus, on the magenta/pink side.

- With both endpoints exactly on the locus (Duv 0), the dip is **Duv ≈ −0.0064 at 4000 K**. This is also the deepest point on the line, at about 3900 K, reached with about 51 % of the luminance from the 6500 K channel.
- Other points on the same line:

CCT	Duv
3000 K	−0.0034
3500 K	−0.0059
4500 K	−0.0058
5000 K	−0.0046
5700 K	−0.0026

- With a typical 6500 K LED at ANSI-nominal Duv +0.003, the dip at 4000 K becomes −0.0049.
- The ANSI C78.377 nominal at 4000 K is +0.001 with a tolerance of ±0.006 [memory], so the uncorrected mix sits at or just outside the tolerance band.
- About 2.5–5 % green, by luminance, brings the mix back onto the locus (see the table in §1.3). This changes mel DER by less than 0.01.
- People often *prefer* a white slightly below the locus (Ohno & Fein 2014, around −0.005 to −0.01 [memory]), so a small residual negative Duv is acceptable. Calibrate against each LED's measured bin chromaticity rather than its nominal value.

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2. Moon texture data: NASA SVS "CGI Moon Kit" (ID 4720)

The page is https://svs.gsfc.nasa.gov/4720. It was first published 6 Sep 2019 and last updated 9 Jan 2026. Visualizer: Ernie Wright (USRA); scientist: Noah Petro (NASA/GSFC).

All files share the base URL `https://svs.gsfc.nasa.gov/vis/a000000/a004700/a004720/`. Every URL below returned HTTP 200 to a HEAD request; byte counts are the content-length header [verified].

Projection.

- The maps use a simple cylindrical (equirectangular) projection with a 2:1 aspect ratio.
- They are "centered on 0° longitude", so the near side is in the middle. The top row is 90° N (TIFF orientation upper-left).

2.1 Colour (albedo) maps

2025 map. It was built from the LROC WAC Hapke-normalised mosaic, using bands 643, 566 and 415 nm for R, G and B. It preserves more colour and dynamic range than the 2019 map and has more polar detail.

File	Pixels	Size
`lroc_color_2k.jpg`	2048×1024	457,942 B
`lroc_color_16bit_srgb_4k.tif`	4096×2048	61.9 MB
`lroc_color_16bit_srgb_8k.tif`	8192×4096	232 MB
`lroc_color_16bit_srgb_16k.tif`	16384×8192	909 MB
`lroc_color_16bit_srgb.tif`	full size	2.4 GB
`lroc_color.exr`	full size, float16, linear	943 MB

2019 map. 24-bit RGB.

File	Pixels	Size
`lroc_color_poles_1k.jpg`	1024×512	139,068 B
`lroc_color_poles_2k.tif`	2048×1024	3,339,438 B
`lroc_color_poles_4k.tif`	4096×2048	13,095,388 B
`lroc_color_poles_8k.tif`	8192×4096	48.3 MB
`lroc_color_poles_hw5x3.tif`	9600×3240	49.1 MB
`lroc_color_poles_16k.tif`	16384×8192	178 MB
`lroc_color_poles.tif`	27360×13680	494 MB

Coverage caveat: the source mosaic covers only 70° N to 70° S. The poles are filled in from the lower-resolution LOLA albedo map (LDAM), which makes up more than 20 % of the pixels but only 6 % of the surface. NASA notes the map is "optimized for aesthetics, not science".

2.2 Displacement (elevation) maps, from LOLA gridded data (spring 2019)

Units, as stated on the page:

- **Float TIFFs** are in **kilometres relative to a 1737.4 km sphere**. They are the source half-metre values divided by 2000.
- **uint16 TIFFs** are in **half-metres, offset by +20 000**: h_km = (DN − 20000) / 2000.

The page writes the uint16 reference radius as "1727400 m". That looks like a typo for 1737400 m.

File	Pixels / resolution	Size
`ldem_3_8bit.jpg`	1024×512 (about 2.8 px/°), 8-bit greyscale	111,552 B
`ldem_4.tif`	1440×720 (4 px/°), float32 km	4,153,352 B
`ldem_4_uint.tif`	1440×720 (4 px/°), uint16	2,076,866 B
`ldem_16.tif`	5760×2880 (16 px/°), float	66.4 MB
`ldem_16_uint.tif`	5760×2880 (16 px/°), uint16	33,201,026 B (31.7 MB)
`ldem_hw5x3.tif`	9600×3240	17,149,748 B
`ldem_64.tif`	23040×11520 (64 px/°), float	1012.6 MB
`ldem_64_uint.tif`	23040×11520 (64 px/°), uint16	506 MB

Vertical scale [computed].

- `ldem_4.tif` spans **−8.879 to +10.504 km**, a range of 19.38 km. The uint16 version decodes to the same values to within 1e-6.
- The **8-bit JPEG's scale is not documented** on the page. A regression against `ldem_4.tif` (r = 0.9965) gives **km ≈ 0.0764 × grey − 8.96**. That means grey 0 is about −9.0 km, grey 255 is about +10.5 km, and each grey level is about 76 m. The file's actual grey values run from 5 to 250.

Scale check for a 50 mm moon [computed]. The full relief range is 1.1 % of the radius, so at 25 mm radius it is only **0.28 mm**, about one layer. True-scale relief is invisible. The lithophane has to map albedo, or exaggerated or high-passed elevation, onto wall thickness.

Resolution check for a 50 mm moon. The equator is 157 mm round.

- The 1k map gives 0.15 mm per pixel at the equator, and the 2k map 0.077 mm per pixel.
- `ldem_4` gives 0.11 mm per pixel.
- A 0.4 mm nozzle resolves about 0.4 mm, so the **2k colour map plus `ldem_4` is more than enough**.

2.3 Files downloaded

They are in `/tmp/claude-0/-home-user/7ba8e87c-d4ee-5a66-9891-51bde4e0a9a7/scratchpad/research/moon/`. Dimensions were checked with PIL.

File	Bytes	Pixels	Mode
lroc_color_2k.jpg	457,942	2048×1024	RGB 8-bit
lroc_color_poles_1k.jpg	139,068	1024×512	RGB 8-bit
lroc_color_poles_2k.tif	3,339,438	2048×1024	RGB 8-bit, LZW
ldem_3_8bit.jpg	111,552	1024×512	L 8-bit
ldem_4.tif	4,153,352	1440×720	float32 (km)
ldem_4_uint.tif	2,076,866	1440×720	uint16

2.4 Usage terms [verified]

- **SVS FAQ:** "All of our content is in the public domain (unless otherwise noted)… free to download, use, and redistribute for whatever purposes you see fit." The exception is licensed music.
- **Credit line requested on page 4720:** "NASA's Scientific Visualization Studio". The datasets used are LRO LOLA DEM and the LROC WAC colour mosaic (ASU).
- **NASA media guidelines:**
  - NASA content is "generally… not subject to copyright in the United States".
  - "NASA should be acknowledged as the source".
  - Commercial use "must not explicitly or implicitly convey NASA's endorsement".
  - The NASA insignia, logotype and identifiers "are not in the public domain".
  - Commercial use "on merchandise or products" is referred to the Merchandise Guidelines (https://www.nasa.gov/nasa-brand-center/merchandise-approvals/).
- **Recommendation:**
  - Credit the data as "Moon texture: NASA's Scientific Visualization Studio, CGI Moon Kit (LRO LROC/LOLA data)" in the documentation and listing.
  - Use no NASA logo, and don't use the NASA name as branding.

    - If the lamp is sold, read the Merchandise Guidelines page first.

Related: SVS 14959, "Moon 3D Models for Web, AR, and Animation" (29 Jan 2026), offers GLB and USDZ models. Examples are `moon_small.glb` (13.2 MB) and `Moon_NASA_LRO_8k_Topo_Small.glb` (77.7 MB).

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3. Thingiverse survey: NOT COMPLETED

What blocked it.

- The shared per-turn WebSearch budget (200 calls across all agents) was used up before any `allowed_domains: ["thingiverse.com"]` search ran.
- A single plain `curl` to `https://www.thingiverse.com/` returned **HTTP 403**, a Cloudflare challenge.
- Printables and MakerWorld homepages also returned 403.
- Following the brief, I did not bypass any of this, and I did not query any search engine or site search another way.
- **No thing IDs, titles, authors or licences are listed here**, because none could be verified. I did not want to invent IDs.

Follow-up plan for a fresh session.

- Run WebSearch with `allowed_domains: ["thingiverse.com"]` for each of these queries:
    - "moon lamp"
    - "moon lithophane"
    - "lithophane moon lamp LED"
    - "moon lamp 60mm"
    - "small moon lamp"
    - "moon night light"
- For each result, record: thing ID, title, author, licence, diameter, minimum and maximum wall thickness, and the hole diameter or base.
- If the CC licence isn't shown in the snippet, record it as unknown.

Non-Thingiverse generators, checked directly this session [verified].

- **3dp.rocks/lithophane** (open source, github.com/MarkDurbin104/3dp.rocks). Defaults from `data/layout.json`:
    - Thickness 3 mm, Thinnest Layer 0.8 mm, Vectors per Pixel 4, Max Size 100 mm, Border 0.
    - Forms: flat, inner/outer curve, cylinder, pillow, dome (image on top or on side), heart. There is **no sphere form**.
    - On commercial use: "Yes, you can use the models to make money or in any commercial setting".
- **lithophanemaker.com** has Sphere and Globe types, plus a "sphere_moon_background" option. Defaults, read from its JS bundle rather than its documentation:
    - Sphere diameter 140 mm; lithophane min/max 0.8 / 2.8 mm.
    - Bottom cylinder 70 mm OD, 4 mm thick, 15 mm high.
    - Optional light base with a 40 mm ID light shaft, a 20×20 mm cable hole, and 45° maximum overhang.
- **ItsLitho** offers a Sphere shape among Plane, Cylinder, Arc, Pumpkin, Vases, Bell and Lamp.

Lessons for the 50 mm design (from the generators and guides in §4):

- Common practice is min ≈ 0.8 mm and max ≈ 2.8–3.6 mm, printed fully solid.
- Generator spheres are 100–140 mm with 40–70 mm openings. Scaling one down to 50 mm would leave a 0.3–1 mm wall and a 14–25 mm hole. That supports the team's decision to generate its own geometry at full thickness.

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4. Lithophane print practice

Parameter	Recommendation	Source
Min / max thickness, white PLA	**0.8 / 3.0 mm** typical. eSUN PLA+ Cool White 0.8 / 3.0–3.2; SUNLU White 0.8 / 2.8; Tinmorry White 0.8 / 3.6	ItsLitho filament pages; 3dp.rocks defaults 0.8 / 3.0; lithophanemaker 0.8 / 2.8 [verified]
Calibration	Print a calibration (step) test with each filament before the real part, because each has its own "sweet spot"	ItsLitho [verified]
Fill	**100 %** infill (ItsLitho uses 99 % to save time), *or* many walls (e.g. 99) with 0 % infill so the lines follow the contours. Bambu: "100% Rectilinear"	ItsLitho; Bambu wiki [verified]
Layer height	**0.12 mm** is ItsLitho's sweet spot; 0.2 mm is the coarsest acceptable. Bambu CMYK litho uses a 0.15 mm first layer and a 0.2 mm nozzle	ItsLitho; Bambu wiki [verified]
Speed	**30 mm/s** best; 45 mm/s is acceptable	ItsLitho [verified]
Orientation	Flat lithophanes print vertical, because XY resolution is better than Z steps. Spheres and globes need supports touching the build plate only: 60° overhang threshold, concentric pattern	ItsLitho [verified]
Material	White PLA is the standard (Bambu mono litho: PLA Basic Jade White). **PETG:** min 0.8 mm, layers ≥ 0.16 mm, 250–260 °C, little fan; better outdoors and in heat. Resin: 0.6 / 3.8–4 mm	ItsLitho PETG/resin post (community author) [verified]
Heat	Avoid hot light sources: "really powerful LEDs can get pretty hot… especially with PLA". PLA softens at about 55–60 °C [memory], so keep LED power and the thermal path in check inside a sealed 50 mm sphere	Prusa blog [verified]

Light transmission and reflectance of white PLA against thickness: no reliable measured data found.

- I looked for percent transmission at 0.6, 1, 2 and 3 mm, and for reflectance or absorption figures. None turned up before the search budget ran out.
- No Prusa lithophane guide exists in the Prusa help or blog sitemaps; only an LED-lamp post turned up.
- The Bambu wiki has CMYK and backlight-kit pages but no transmission data.
- HueForge's per-filament "Transmission Distance (TD)" is the closest community dataset. It was not retrieved.

Recommended in-house measurement:

- Print a step wedge at 0.6, 0.8, 1.0, 1.4, 1.8, 2.2, 2.6, 3.0 and 3.6 mm, using the production filament, layer height and line width.
- Put it over each LED channel (2700 K, 6500 K, R, G, B) in a dark box and measure with a lux meter or spectrometer.

- This gives the transmission curve, and also any colour shift with thickness. White pigment (TiO_2) may warm the light passing through thick areas; this is a hypothesis to test.

Sources

Circadian, melanopic and light modes

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